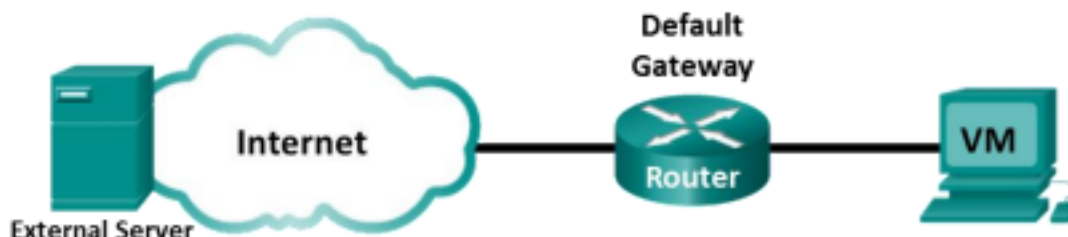


Lab - Exploring Nmap

Topology



Objectives

Part 1: Exploring Nmap

Part 2: Scanning for Open Ports

Background / Scenario

Port scanning is usually part of a reconnaissance attack. There are a variety of port scanning methods that can be used. We will explore how to use the Nmap utility. Nmap is a powerful network utility that is used for network discovery and security auditing.

Required Resources

- CyberOps Workstation virtual machine
- Internet access

Instructions

Part 1: Exploring Nmap

In this part, you will use manual pages (or man pages for short) to learn more about Nmap.

The **man** [*program* | *utility* | *function*] command displays the manual pages associated with the arguments. The manual pages are the reference manuals found on Unix and Linux OSs. These pages can include these sections: Name, Synopsis, Descriptions, Examples, and See Also.

- Start CyberOps Workstation VM.
- Open a terminal.
- At the terminal prompt, enter **man nmap**.

```
[analyst@secOps ~]$ man nmap
```

What is Nmap?

Nmap (Network Mapper) là một công cụ mã nguồn mở dùng để phát hiện mạng và kiểm tra an ninh mạng.

What is nmap used for?

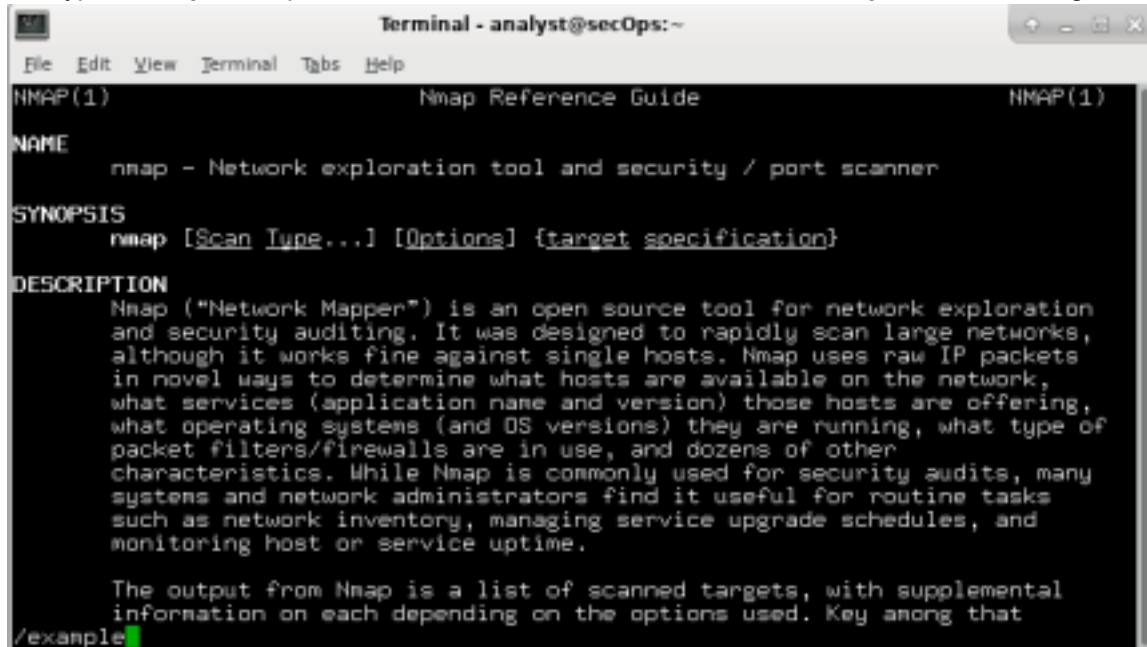
Nmap được sử dụng để khám phá các thiết bị/máy chủ đang hoạt động trên mạng, quét các cổng mở (open ports), phát hiện hệ điều hành, xác định các dịch vụ đang chạy cùng phiên bản của chúng, và đánh giá mức độ bảo mật của hệ thống.

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- d. While in the man page, you can use the up and down arrow keys to scroll through the pages. You can also press the space bar to forward one page at a time.

To search for a specific term or phrase use enter a forward slash (/) or question mark (?) followed by the term or phrase. The forward slash searches forward through the document, and the question mark searches backward through the document. The key **n** moves to the next match.

Type **/example** and press ENTER. This will search for the word **example** forward through the man page.

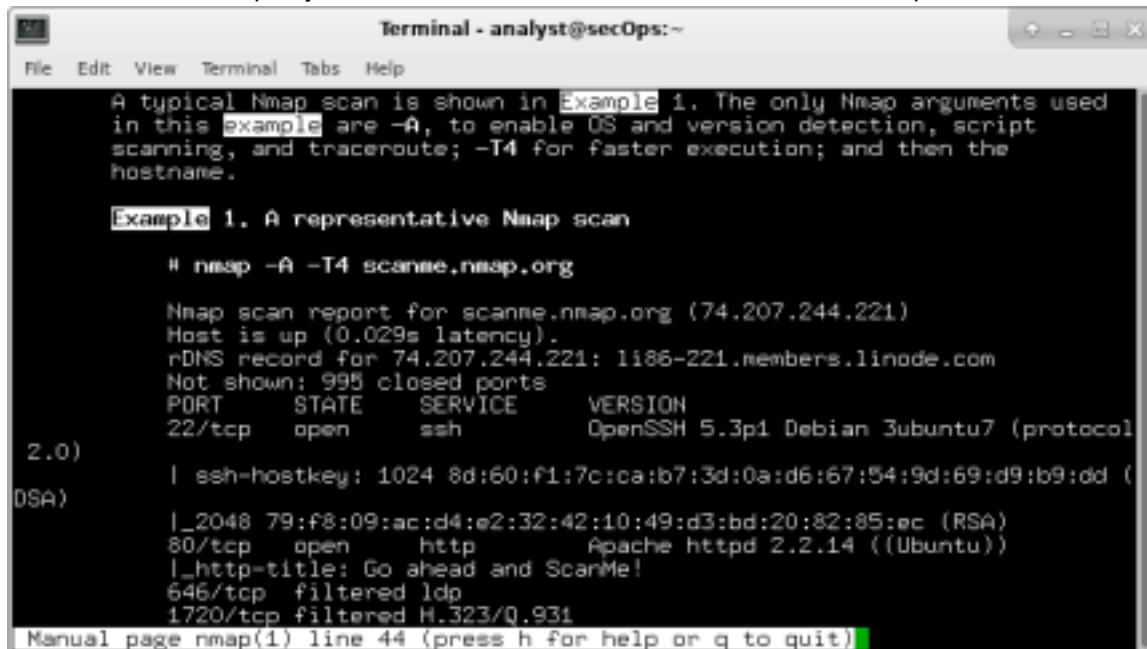


```

Terminal - analyst@secOps:~
File Edit View Terminal Tabs Help
NMAP(1)                                Nmap Reference Guide                                NMAP(1)
NAME
    nmap - Network exploration tool and security / port scanner
SYNOPSIS
    nmap [Scan Type...] [Options] {target specification}
DESCRIPTION
    Nmap ("Network Mapper") is an open source tool for network exploration
    and security auditing. It was designed to rapidly scan large networks,
    although it works fine against single hosts. Nmap uses raw IP packets
    in novel ways to determine what hosts are available on the network,
    what services (application name and version) those hosts are offering,
    what operating systems (and OS versions) they are running, what type of
    packet filters/firewalls are in use, and dozens of other characteristics.
    While Nmap is commonly used for security audits, many systems and network
    administrators find it useful for routine tasks such as network inventory,
    managing service upgrade schedules, and monitoring host or service uptime.

    The output from Nmap is a list of scanned targets, with supplemental
    information on each depending on the options used. Key among that
/example
  
```

- e. In the first instance of example, you see three matches. To move to the next match, press **n**.



```

Terminal - analyst@secOps:~
File Edit View Terminal Tabs Help
A typical Nmap scan is shown in Example 1. The only Nmap arguments used
in this example are -A, to enable OS and version detection, script
scanning, and traceroute; -T4 for faster execution; and then the
hostname.

Example 1. A representative Nmap scan

# nmap -A -T4 scanme.nmap.org

Nmap scan report for scanme.nmap.org (74.207.244.221)
Host is up (0.029s latency).
rDNS record for 74.207.244.221: li86-221.members.linode.com
Not shown: 995 closed ports
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 5.3p1 Debian 3ubuntu7 (protocol
2.0)
| ssh-hostkey: 1024 8d:60:f1:7c:ca:b7:3d:0a:d6:67:54:9d:69:d9:b9:dd (
DSA)
|_ 2048 79:f8:09:ac:d4:e2:32:42:10:49:d3:bd:20:82:85:ec (RSA)
80/tcp    open      http         Apache httpd 2.2.14 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
646/tcp   filtered  ldp
1720/tcp  filtered  H.323/Q.931

Manual page nmap(1) line 44 (press h for help or q to quit)
  
```

Example 1.

Look at

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What is the **nmap** command used?

Cú pháp lệnh được sử dụng trong Ví dụ 1 là: **nmap -A -T4 scanme.nmap.org**

Use the search function to answer the following questions.

What does the switch -A do?

Cờ -A bật các tính năng dò tìm nâng cao, bao gồm: phát hiện hệ điều hành (OS detection), phát hiện phiên bản dịch vụ (version detection), chạy các kịch bản kiểm tra (script scanning) và xác định tuyến đường (traceroute).

What does the switch -T4 do?

Cờ -T4 thiết lập tốc độ quét theo mức thời gian ngầm định là "Aggressive" (nhanh và linh hoạt hơn), giúp rút ngắn thời gian quét mà vẫn đảm bảo độ tin cậy trên các mạng tương đối ổn định.

f. Scroll through the page to learn more about nmap. Type **q** when finished.

Part 2: Scanning for Open Ports

In this part, you will use the switches from the example in the Nmap man pages to scan your localhost, your local network, and a remote server at scanme.nmap.org.

Step 1: Scan your localhost.

a. If necessary, open a terminal on the VM. At the prompt, enter **nmap -A -T4 localhost**. Depending on your local network and devices, the scan will take anywhere from a few seconds to a few minutes.

```
[analyst@secOps ~]$ nmap -A -T4 localhost
```

```
Starting Nmap 7.40 ( https://nmap.org ) at 2017-05-01 17:20 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000056s latency).
Other addresses for localhost (not scanned): ::1
rDNS record for 127.0.0.1: localhost.localdomain
Not shown: 996 closed ports
PORT STATE SERVICE VERSION
21/tcp open  ftp vsftpd 2.0.8 or later
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ -rw-r--r-- 1 0 0 0 Apr 19 15:23 ftp_test
<some output omitted>
```

b. Review the results and answer the following questions.

Which ports and services are opened?

- Cổng 21/tcp: Dịch vụ **FTP**.
- Cổng 22/tcp: Dịch vụ **SSH**.
- Cổng 23/tcp: Dịch vụ **Telnet**.
- Cổng 80/tcp: Dịch vụ **HTTP**.

For each of the open ports, record the software that is providing the services.

- Cổng 21 (FTP): Phần mềm **vsftpd 2.0.8** (hoặc mới hơn).
- Cổng 22 (SSH): Phần mềm **OpenSSH 8.2** (protocol 2.0).
- Cổng 23 (Telnet): Phần mềm **Openwall GNU*/Linux telnetd**.
- Cổng 80 (HTTP): Phần mềm **Apache httpd**.

Step 2: Scan your network.

Warning: Before using Nmap on any network, please gain the permission of the network owners before proceeding.

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- a. At the terminal command prompt, enter **ip address** to determine the IP address and subnet mask for this host. For this example, the IP address for this VM is 10.0.2.15 and the subnet mask is 255.255.255.0.

```
[analyst@secOps ~]$ ip address
<output omitted>
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:ed:af:2c brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85777sec preferred_lft 85777sec
    inet6 fe80::a00:27ff:feed:af2c/64 scope link
        valid_lft forever preferred_lft forever
```

Record the IP address and subnet mask for your VM.

Which network does your VM belong to?

Máy ảo thuộc dải mạng 10.0.2.0/24.

- b. To locate other hosts on this LAN, enter **nmap -A -T4 network address/prefix**. The last octet of the IP address should be replaced with a zero. For example, in the IP address 10.0.2.15, the .15 is the last octet. Therefore, the network address is 10.0.2.0. The /24 is called the prefix and is a shorthand for the netmask 255.255.255.0. If your VM has a different netmask, search the internet for a “CIDR conversion table” to find your prefix. For example, 255.255.0.0 would be /16. The network address 10.0.2.0/24 is used in this example

Note: This operation can take some time, especially if you have many devices attached to the network. In one test environment, the scan took about 4 minutes.

```
[analyst@secOps ~]$ nmap -A -T4 10.0.2.0/24

Starting Nmap 7.40 ( https://nmap.org ) at 2017-05-01 17:13 EDT
<output omitted>
Nmap scan report for 10.0.2.15
Host is up (0.00019s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r-- 1 0 0 0 Mar 26 2018 ftp_test
| ftp-syst:
| STAT:
| FTP server status:
| Connected to 10.0.2.15
| Logged in as ftp
| TYPE: ASCII
```

```
| No session bandwidth limit
| Session timeout in seconds is 300
| Control connection is plain text
| Data connections will be plain text
| At session startup, client count was 1
| vsFTPD 3.0.3 - secure, fast, stable
|_End of status
```

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```
22/tcp open ssh OpenSSH 8.2 (protocol 2.0)
23/tcp open telnet Openwall GNU/*/Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Post-scan script results:

```
| clock-skew:
| 0s:
| 10.0.2.4
| 10.0.2.3
|_ 10.0.2.2
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 256 IP addresses (4 hosts up) scanned in 346.89 seconds

How many hosts are up?

Có 4 hosts đang hoạt động (hosts up).

From your Nmap results, list the IP addresses of the hosts that are on the same LAN as your VM. List some of the services that are available on the detected hosts.

- Các địa chỉ IP tìm thấy: 10.0.2.2, 10.0.2.3, 10.0.2.4, 10.0.2.15.
- Dịch vụ khả dụng trên các máy chủ: Dịch vụ FTP (cổng 21), SSH (cổng 22), Telnet (cổng 23), HTTP (cổng 80), cùng các phản hồi đồng bộ thời gian (clock-skew).

Step 3: Scan a remote server.

- a. Open a web browser and navigate to **scanme.nmap.org**. Please read the message posted.

What is the purpose of this site?

Trang web này do dự án Nmap dựng lên nhằm mục đích cho phép người dùng thử nghiệm và học cách sử dụng công cụ Nmap mà không vi phạm pháp luật hay xâm phạm trái phép vào các hệ thống khác.

- b. At the terminal prompt, enter **nmap -A -T4 scanme.nmap.org**.

```
[analyst@secOps Desktop]$ nmap -A -T4 scanme.nmap.org
```

```
Starting Nmap 7.40 ( https://nmap.org ) at 2017-05-01 16:46 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.040s latency).
Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f Not shown: 992 closed ports
PORT STATE SERVICE VERSION
```

```
22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.8 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_ 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
25/tcp filtered smtp
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp filtered msrpc
139/tcp filtered netbios-ssn
```

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```
445/tcp filtered microsoft-ds
593/tcp filtered http-rpc-epmap
4444/tcp filtered krb524
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Service detection performed. Please report any incorrect results at
<https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 23.96 seconds

c. Review the results and answer the following questions.

Which ports and services are opened?

- Cổng **22/tcp**: Dịch vụ **SSH** (OpenSSH 6.6.1p1 Ubuntu).
- Cổng **80/tcp**: Dịch vụ **HTTP** (Apache httpd 2.4.7).
- Cổng **9929/tcp**: Dịch vụ **nping-echo** (Nping echo).
- Cổng **31337/tcp**: Dịch vụ **tcpwrapped**.

Which ports and services are filtered?

Các cổng bị lọc (filtered) bởi tường lửa bao gồm:

- Cổng **25/tcp**: smtp
- Cổng **135/tcp**: msrpc
- Cổng **139/tcp**: netbios-ssn
- Cổng **445/tcp**: microsoft-ds
- Cổng **593/tcp**: http-rpc-epmap
- Cổng **4444/tcp**: krb524

What is the IP address of the server?

Địa chỉ IP của máy chủ là **45.33.32.156**

What is the operating system?

Hệ điều hành được phát hiện là **Linux** (Linux kernel).

Reflection Question

Nmap is a powerful tool for network exploration and management. How can Nmap help with network security?
How can Nmap be used by a threat actor as a nefarious tool?

